AUTOMATION PROJECT DOCUMENTATION

LinkedIn Auto-Post Agent

Setup, Configuration & Operations Manual

A practical guide to importing, configuring, testing, and maintaining the n8n workflow.

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n8n + Google Gemini + Cloudflare Workers AI + Gmail + LinkedIn + Google Sheets

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How to use this manual

Read Chapters 1-3 for the architecture and requirements, then follow Chapters 4-6 to import the JSON and connect accounts. Use Chapters 8-11 for testing, troubleshooting, security, and release checks.

Chapter 1 | Project Overview

The LinkedIn Auto-Post Agent is a semi-automated n8n workflow. It turns a short note about something learned into a draft LinkedIn post and a generated image, emails a preview, waits for an explicit approval, then publishes the approved content and records a log row in Google Sheets.

The workflow is designed to keep a person in control of publishing. It does not publish just because the AI generated a post: the user must choose “Post it” in the approval email. Choosing “Skip”, or allowing the approval to time out, ends the run without publishing.

What the workflow is intended to do

- Collect the topic, the user’s notes, and an intended audience through an n8n form.
- Use Google Gemini to draft a first-person LinkedIn post from the supplied context and notes.
- Use a second Gemini agent to create a photo-style image prompt based on the post.
- Generate an image through Cloudflare Workers AI using the FLUX.1 schnell model endpoint.
- Send an image preview email and a separate approval email.
- Publish the post with an image to LinkedIn after approval, then append a log row to Google Sheets.

What this manual covers

This manual focuses on the supplied exported workflow JSON and its companion README, workflow diagram, setup reference PDF, and privacy-policy HTML. Where those files disagree, the manual calls out the discrepancy instead of pretending the current workflow already implements a feature that is only planned.

Important distinction
“Semi-automated” is the accurate description: content generation and image creation are automated, but publishing requires an approval action. The workflow runs on the n8n instance where the user imports and configures it.

Who should use this guide

The reader should be comfortable opening an n8n workflow, creating credentials, copying API keys into credential forms, editing a Google Sheet, and performing a manual test run. A beginner can follow the steps, but OAuth and provider dashboards may require careful configuration.

Chapter 2 | Workflow Architecture

The supplied diagram shows the main process in five visual stages: Input, Content Processing, Image Generation, Approval Flow, and Publish & Track. The diagram is a high-level map; node settings and current implementation limitations are explained in the following chapters.

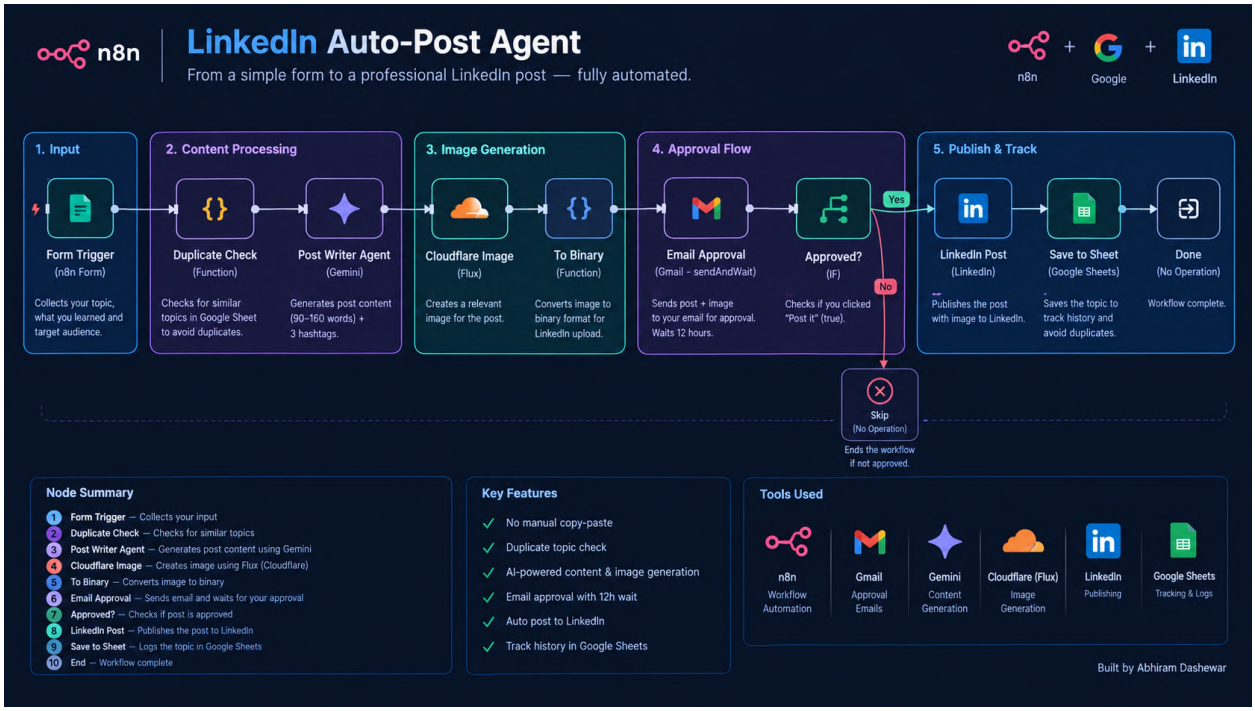


Figure 2.1. Workflow overview supplied with the project files. Use the node reference in Chapter 7 for implementation details.

End-to-end data flow

Stage	What happens	Main output
1. Input	The form receives Topic, What I learned, and Who is this for?	Validated topic, audience, and notes
2. Content processing	The workflow pauses, then Gemini writes the post in the configured voice.	Plain-text LinkedIn post
3. Image generation	Gemini writes a descriptive image prompt; Cloudflare generates the image; the Code node converts it into binary image data.	Post text plus image file
4. Approval	Gmail sends an image preview, then sends Post it / Skip approval and waits for a response.	Approval result
5. Publish and track	Approved runs restore the image, publish to LinkedIn, then append a row to Google Sheets.	Published post and log row

Execution branches

The “Approved?1” IF node checks whether the approval result is true. The true branch continues to “Restore Image1” and then “Post to LinkedIn”. The false/no-approval outcome does not lead to the publishing branch. The current exported connections show the positive path connected onward; the alternative path simply ends.

Current implementation note
The overview diagram labels the early Code node “Duplicate Check”, but the exported code only trims and validates the two required form fields. It does not query the Google Sheet or stop an already-used topic. Do not rely on this workflow to prevent duplicate topics until that logic is added and tested.

Chapter 3 | Prerequisites and Environment

Required components

Component	Purpose	What to prepare
n8n self-hosted	Runs the workflow. The owner's Docker setup can be used.	A working n8n instance with access to the editor and credential settings.
Google Gemini API	Drafts the LinkedIn post and the image prompt.	An API key and an available model configured in the Gemini node credential.
Cloudflare Workers AI	Generates the image using FLUX.1 schnell.	Cloudflare account ID and a Workers AI API token with the required permission.
Gmail / Google account	Sends the image preview and approval email.	An authorized Gmail credential configured in the two Gmail nodes.
Google Sheets	Stores post history and execution details.	A spreadsheet with a tab named Posts, the expected columns, and an authorized Sheets credential.
LinkedIn Developer app	Allows the workflow to create a post for the authorized person.	A LinkedIn account, developer app, enabled share/post product or permission, and connected n8n credential.

About self-hosting with Docker

- Keep n8n running when scheduled or form-triggered executions need to be processed. For this workflow, a user normally submits the form manually, so n8n must be available at submission and while the run waits for approval.
- Keep the n8n persistent data volume and encryption key protected. Credentials are stored by n8n, and losing or changing the encryption key can make saved credentials unusable.
- Use a stable, secure HTTPS address if your OAuth provider or any future webhook needs to redirect back to your n8n host. A local-only URL may work for some local flows but will not be reachable from external services.
- Do not expose the n8n editor to the public internet without authentication, HTTPS, updates, and an appropriate deployment configuration.

Minimum readiness check

- You can open your n8n editor and run a simple test workflow.
- You can create and save credentials in n8n.
- Your Google Sheet is accessible to the Google account used in the credential.
- You can receive email at the address where approvals will be sent.
- You can authorize a LinkedIn credential with permission to publish a post as the account owner.

Provider plans and quotas
This project was designed around available provider APIs and includes pauses between AI calls. Free quotas and model availability can change. Treat "free" as dependent on each provider's current account, region, quota, and pricing terms, not as a guarantee.

Chapter 4 | Import the Workflow and Configure It

Step 1. Import the JSON

- Open your own n8n editor.
- Create or open the Workflows area and choose the import-from-file option. The label may vary slightly by n8n version.
- Select the sanitized workflow JSON file supplied with the repository.
- Review the imported canvas and ensure all 18 nodes appear. Save the workflow under a recognizable name such as LinkedIn Auto-Post Agent - Local Copy.

Step 2. Understand the Config node

The Set node named Config centralizes the main settings that are referenced later in the workflow. Replace the template values in your own local copy. Keep personal profile context professional and avoid entering private or sensitive personal details.

Field	What it controls	Example / instruction
sheetId	Google Sheets spreadsheet identifier passed to the logging node.	Copy the ID portion from the spreadsheet URL, not the entire URL.
niche	A description of the broad subject area.	For example: “cybersecurity basics and electronics learning”. The current exported prompts do not visibly reference this field directly, so do not assume it changes generated output unless you wire it into a prompt.
myProfile	Background, skills, current learning stage, voice preferences.	Use only facts you are comfortable sending to the configured AI provider.
myGoals	Audience, content goals, and intended professional direction.	Keep it accurate and avoid inventing experience or achievements.
imageStyle	A short visual direction for the image prompt.	Example: realistic student workspace, natural light, notebook and hardware, no logos or readable text.
cloudflareAccountId	Account portion of the Cloudflare Workers AI endpoint.	Use the account ID from your own Cloudflare dashboard. It is an identifier, not the API token.

Step 3. Review the form

The form is titled “What did you learn today?” and has three fields: Topic (required), What I learned (required textarea), and Who is this for? (dropdown). The default audience options are Students and beginners, Aspiring cybersecurity learners, and Recruiters and professionals. The form path is generated by n8n; after import, use the URL shown by your own Form Trigger node rather than copying a path from another installation.

Step 4. Review the two Gemini agent nodes

Assign the appropriate Google Gemini credential to both Gemini language-model nodes. The README and setup reference show a Gemini model identifier, but model availability and naming may differ by account or n8n version. Select a model actually available to your API key and test the nodes independently. The writer temperature in the exported node is 0.7, and the image-prompt model temperature is 0.9.

Expression pre-flight check
In the exported JSON, the Image Prompt Agent text and system message begin with a double equals sign (==), unlike the usual single leading equals sign used for an n8n expression. After import, inspect those two fields in the editor and test the node. If n8n reports an expression or syntax error, correct the leading marker to the syntax expected by your installed n8n version before running the entire workflow.

Chapter 5 | Prepare the Google Sheet

Create a Google spreadsheet you control, then add a worksheet tab named exactly Posts. Row 1 should contain the log column headers. The Save to Google Sheet node uses append-row behavior and does not read rows before deciding whether to publish.

Recommended header row

Column header	What is stored
Date	Date and time of the logging step, formatted as yyyy-MM-dd HH:mm.
Topic	Topic submitted in the form.
Audience	Selected or default target audience.
Post	Generated LinkedIn post text.
ImagePrompt	Prompt produced by the Image Prompt Agent.
LinkedInPostId	Post identifier returned by LinkedIn when available.

Important mapping mismatch in the supplied JSON

The exported Save to Google Sheet node currently maps two fields using the names “Auidence” and “Post ” (with a trailing space). The README’s proposed header row uses “Audience” and “Post”. This mismatch may cause missing or incorrectly matched values, depending on the node schema and sheet headers. Before publishing the template, standardize the node mapping and schema to “Audience” and “Post”, refresh the column mapping in n8n, and make the sheet headers match exactly.

Recommended fix
Use the six headers shown above in row 1. In the Google Sheets node, correct “Auidence” to “Audience” and remove the trailing space from “Post ”. Then re-select or refresh the sheet columns in the node and test appending a sample row.

Connect the spreadsheet

- Authorize the Google Sheets credential inside n8n using the Google account that can edit the spreadsheet.
- Paste your own spreadsheet ID into Config.sheetId.
- Confirm that the worksheet tab name is Posts, including capitalization.
- Use the node’s test/run option on a test copy of the sheet before enabling real publishing.
- Do not publish your private spreadsheet, its URL, or data rows in a public repository unless you have deliberately made them public.

What the sheet does not do yet

The sheet is an append-only history log in the supplied workflow. It does not implement duplicate detection: the exported workflow contains no Google Sheets read/search node before content generation or publishing. If duplicate prevention is a requirement, add a read step and a clear comparison rule, then branch away from the publishing path when a duplicate is found. Test that change separately.

Chapter 6 | Configure Credentials and Connected Services

Credentials are not portable with a workflow export. Each user must connect their own accounts in n8n. The exact provider dashboard labels can change, so use this chapter as a logical checklist and follow the current provider interface when creating keys or OAuth applications.

6.1 Google Gemini API

- Create or select a project/account in Google AI Studio and create an API key using the provider's current instructions.
- Create or open the Google Gemini credential in n8n and enter the key. Do not paste the key into the Config node or README.
- Assign that credential to both the Gemini - Writer and Gemini - Image Prompt language model nodes. The exported JSON appears to include a credential reference on the image-prompt model node only; confirm the writer model is also connected after import.
- Run each model node separately before executing the full workflow. Rate limits, quota exhaustion, and model availability can stop the run.

6.2 Cloudflare Workers AI

- Use your Cloudflare account ID in Config.cloudflareAccountId.
- Create a Cloudflare API token with the Workers AI permissions required for your account and model use.
- In n8n, create an HTTP Header Auth credential for the Cloudflare Image1 request. The header name is Authorization and its value format is "Bearer YOUR_WORKERS_AI_TOKEN" (the word Bearer, one space, then the token).
- The request uses POST to the account-scoped endpoint ending in /ai/run/@cf/black-forest-labs/flux-1-schnell. The current exported request sends a JSON body with a prompt and steps set to 6, with a 120,000 ms timeout.
- Never use a token intended for a different Cloudflare product, and never commit the token to GitHub.

6.3 Gmail / approval emails

- Connect a Gmail OAuth credential in n8n, or use an email credential supported by your node configuration.
- Assign the same sending account to Email Preview1 and Email Approval1 if that is your intended design.
- Set the recipient address in both nodes to the inbox where you want to review and approve posts. In the sanitized template, this field is YOUR_APPROVAL_EMAIL.
- The preview email includes the generated image attachment and post text. The next email provides Post it and Skip choices and waits for approval for up to 12 units as configured in the node. Confirm the displayed unit in your n8n version.

6.4 LinkedIn publishing

- Use a LinkedIn account and developer application that are authorized for the intended post-publishing capability.
- In n8n, open the LinkedIn credential and complete the OAuth flow using the redirect/callback URL shown by n8n. Register that exact URL in the LinkedIn app settings when required.
- Enter your own developer app credentials where requested and grant only the permissions needed for the workflow. Permission names and approval requirements may change, so follow the current LinkedIn developer documentation.
- Select the intended personal profile identity and confirm that the credential is authorized to publish as that person. Test with a non-sensitive post after approval.

6.5 Google Sheets OAuth

- Create or open the Google Sheets credential in n8n and authorize the Google account that owns or edits the target spreadsheet.
- If you configure a custom OAuth client in Google Cloud, copy the redirect URI exactly as shown by n8n into the Google OAuth client's authorized redirect URIs. Use the correct application type and provider settings for your environment.
- Client IDs and client secrets are not interchangeable with API keys. If a connection reports invalid_client, check the client ID/secret pair, the OAuth client type, and registered redirect URI, then reconnect the credential.

Secrets safety
Do not include working client secrets, API keys, tokens, or credentials in the workflow JSON, screenshots, PDF, GitHub history, or the privacy-policy repository. If a credential may already have been exposed, revoke or rotate it with the provider before publishing.

OAuth configuration checklist

- When a provider requires a web OAuth client, select the appropriate web application client type and copy the redirect URI exactly as n8n displays it, including the protocol, hostname, port, and path.
- Make sure the client ID and client secret belong to the same OAuth application. Do not paste a LinkedIn secret into a Google credential or vice versa.
- Confirm that the APIs or products needed by the workflow are enabled for the provider account and that the developer application has the requested permissions.
- If the provider requires a consent screen, test-user list, application review, or verification, complete the provider's current setup steps before attempting the OAuth flow.
- After changing OAuth settings, reconnect the n8n credential and test the specific service node before rerunning the full workflow.

Avoiding unnecessary API requests

- Run the two Gemini language-model nodes and the Cloudflare request individually while configuring them.
- Do not repeatedly execute the whole workflow when a single API node is failing. Each run may use quota or incur charges.
- Keep the existing wait nodes unless you have deliberately reviewed rate-limit behavior. Increasing pauses can reduce request bursts, but it cannot guarantee quota availability.
- Read provider error messages before retrying. A quota error, invalid credential, and malformed request need different fixes.

Chapter 6 | Beginner Configuration Reference

Use this reference with Sections 6.1-6.5. It reconciles the older duplicate table against the current 18-node workflow. Where values conflict, follow the current workflow export and the instructions in this manual.

Node	Field / setting	Value and beginner instructions
1. Form Trigger1	Form Path	Automatically generated by n8n. After import, open this node and use the form URL shown by your own instance. Do not rely on the example UUID in the older reference PDF.
2. Config	sheetId	Paste the spreadsheet ID only, not the full URL. It is the text between **/d/** and **/edit** in the Google Sheets URL. Create a worksheet tab named exactly **Posts**.
2. Config	niche	Describe your broad topic, for example *cybersecurity basics and electronics learning*. The current prompt expressions may not use this field directly, so changing it may not affect the generated post by itself.
2. Config	myProfile	Add accurate background, skills, projects, interests, and writing preferences. Only include details you are comfortable sending to the configured Gemini service.
2. Config	myGoals	Describe career direction, intended audience, and what you want to be known for. Avoid private health, family, financial, and contact details.
2. Config	imageStyle	Describe the visual style you prefer. Example: realistic student workspace, natural light, notebook, laptop, and electronics; no readable text or logos.
2. Config	cloudflareAccountId	Copy the account ID from your Cloudflare dashboard. This identifies your account in the API endpoint. **It is not the API token.**
3. Duplicate Check	Code node	No setup changes normally needed. In the current export it validates required form fields, but does **not** search Google Sheets for duplicate topics.
4. Wait-1	Wait node	Leave the exported delay unchanged unless you deliberately want to adjust execution timing.
5. Post Writer Agent	AI Agent	Leave its main instructions unchanged initially. It uses your submitted notes and profile/goals context to draft the LinkedIn post.
6. Gemini - Writer	Credential, model, temperature	Select a saved Google Gemini API credential in n8n. Choose a model available to your account. The current workflow export sets temperature to **0.7**. Test the node alone.

Starter prompt for myProfile and myGoals

If you use an AI assistant that has relevant context about you, paste this prompt and review the result before adding it to Config: *"Write two separate sections. PROFILE: my background, skills, projects, interests, and writing style in about 300 words. GOALS: my career direction, what I want to be known for, and the audience I want to reach on LinkedIn in about 200 words. Include only information I am comfortable sharing publicly. Exclude health, family, financial, and personal contact details."* If the assistant does not know your background, provide the facts yourself instead.

Keep identifiers and secrets separate

Account ID: identifies a Cloudflare account. **API token:** authorizes the request. **OAuth client ID/secret:** identify an OAuth application. Enter each value only in the field designed for it; never publish live secrets.

Chapter 6 | Node Settings and Credentials

Continued: nodes 7-18

In the table below, “no changes” means keep the imported node settings unless you are deliberately modifying and testing the workflow.

Node	Field / setting	Value and beginner instructions
7. Wait-2	Wait node	No changes normally needed; this is a delay before the image-prompt agent.
8. Image Prompt Agent	AI Agent prompts	Leave the prompt content unchanged initially. The exported JSON contains fields starting with **==**; inspect and test after import, and correct the expression marker only if n8n reports a syntax error.
9. Gemini - Image Prompt	Credential, model, temperature	Use the same Gemini credential as the writer node and select an available model. The current workflow export uses **0.9**. The older duplicate table says 0.7, so follow the current workflow export.
10. Wait-3	Wait node	Leave unchanged unless you intentionally adjust the delay before image generation.
11. Cloudflare Image1	HTTP request and auth	Method: **POST**. The URL expression uses `Config.cloudflareAccountId`. Create an n8n **HTTP Header Auth** credential: header name **Authorization**; value **Bearer YOUR_WORKERS_AI_TOKEN**. There must be one space after Bearer. Use a Workers AI token with required permissions. Do not use the account ID as the token.
12. To Binary	Code node	No changes normally needed. It converts the image response into binary data named `post-image.jpg`. Confirm output contains `binary.data`.
13. Email Preview1	Recipient and Gmail credential	Set `To` to `YOUR_APPROVAL_EMAIL`, replaced with the inbox you use. Connect a working Gmail credential; check that the preview email includes the generated post and image attachment.
14. Email Approval1	Recipient and Gmail credential	Use the intended sending credential and approval inbox. The email offers **Post it** and **Skip**. Confirm the timeout unit shown in your n8n version; do not assume the number 12 means hours.
15. Approved?1	IF node	No changes normally needed. It should route only an approved result to the publish path.
16. Restore Image1	Code node	No changes normally needed. It restores image binary data after the approval step; verify the binary image is present before publishing.
17. Post to LinkedIn	Post settings and OAuth	Resource: **Post**; operation: **Create**; post as: **Person**; media category: **Image**. Set up your own LinkedIn Developer app and n8n LinkedIn credential, complete OAuth using the redirect URL shown by n8n, and authorize the intended personal profile. Keep organization support and legacy options off unless you deliberately need them.
18. Save to Google Sheet	Google Sheets credential and mapping	Connect a Google Sheets credential that can edit your spreadsheet. The Config `sheetId` identifies the spreadsheet; the worksheet tab is **Posts**. Use append-row and map the six expected headers exactly: **Date, Topic, Audience, Post, ImagePrompt, LinkedInPostId**. Do not assume the Gmail credential or OAuth secret automatically configures Sheets.

Quick credential checklist

1. Open the credential selector in the node and create or choose the correct credential type. 2. Enter API keys/tokens only in the credential form, not in the Config node. 3. Save the credential and select it on every node that needs it. 4. Test the individual node before running the full workflow. 5. For OAuth services, use the redirect/callback URL shown by your own n8n instance and confirm the required provider permissions.

Chapter 7 | Node-by-Node Reference

The exported workflow contains 18 nodes. The table below describes their role and the current implementation visible in the supplied JSON. “Duplicate Check” is the existing node label, not a guarantee that duplicate prevention has been implemented.

Node	Type / role	Current behavior
1. Form Trigger1	Form Trigger	Displays the “What did you learn today?” form and collects the topic, learning notes, and audience.
2. Config	Set	Defines sheetId, niche, myProfile, myGoals, imageStyle, and cloudflareAccountId.
3. Duplicate Check	Code	Trims topic and notes; throws an error if either is empty; emits topic, audience, and angle. Does not query Sheets.
4. Wait-1	Wait	Configured with amount 30 before the post-writing agent.
5. Post Writer Agent	AI Agent	Uses form inputs plus profile/goals context to create a first-person post.
6. Gemini - Writer	Gemini chat model	Language model associated with the writer agent; temperature 0.7 in the exported JSON.
7. Wait-2	Wait	Configured with amount 40 before the image-prompt agent.
8. Image Prompt Agent	AI Agent	Creates one photo-style prompt. Current JSON has double leading equals signs in prompt fields; validate after import.
9. Gemini - Image Prompt	Gemini chat model	Language model associated with the image-prompt agent; temperature 0.9 in the exported JSON.
10. Wait-3	Wait	Configured with amount 50 before image generation.
11. Cloudflare Image1	HTTP Request	POSTs the generated prompt to Cloudflare Workers AI FLUX.1 schnell; steps 6 and timeout 120 seconds are configured.
12. To Binary	Code	Converts base64 image output into image binary data named post-image.jpg when needed.
13. Email Preview1	Gmail	Sends the post text and image attachment as an image-preview email.
14. Email Approval1	Gmail Send and Wait	Sends an approval email with Post it / Skip and waits for the decision; timeout amount is configured as 12.
15. Approved?1	IF	Checks whether the approval result is true and controls the publishing path.
16. Restore Image1	Code	Re-attaches binary image data from To Binary after the approval step.
17. Post to LinkedIn	LinkedIn	Creates a personal post and marks the media category as IMAGE.
18. Save to Google Sheet	Google Sheets	Appends date, topic, audience, post text, image prompt, and LinkedIn post ID when available.

Writer-agent rules

- Writes in first person, using simple conversational language and short sentences.
- Uses the profile/goals configuration and the user’s What I learned notes as its context.
- Avoids hype phrases, invented personal stories, unsupported results, and private details.
- Targets 90-160 words, ends with one genuine question, and uses exactly three relevant hashtags, according to the system prompt in the exported JSON.

Image-prompt rules

- Describes a realistic student workspace with details tied to the topic, such as networking equipment, electronics, a laptop, a notebook, or a breadboard.
- Requests one descriptive paragraph of roughly 45-70 words, with framing, lighting, colours, and mood.
- Prohibits readable text, logos, faces, cliché hacker imagery, padlocks, matrix-code visuals, and glossy 3D renders.

Chapter 8 | Run, Review, and Approve a Post

Before the first real run

- Save the workflow after assigning each required credential.
- Confirm that Config has your own sheet ID, profile, goals, image style, and Cloudflare account ID.
- Confirm that the Posts worksheet and its column mapping have been standardized.
- Check both Gemini nodes, the Cloudflare HTTP request, both Gmail nodes, LinkedIn, and Google Sheets for connected credentials.
- Resolve the Image Prompt Agent leading-equals issue if it causes an expression error.
- Use a topic and notes that are safe to send to the configured AI provider. Do not begin with confidential, client, or personal data.

Typical execution

Step	What you do	What you should see
1	Open the form URL shown in Form Trigger1 and submit Topic, What I learned, and the audience.	The run starts and required form values are present.
2	Allow the writer and image-prompt nodes to execute.	A text post and image prompt appear in execution data.
3	Let Cloudflare generate the image.	Cloudflare response is converted into a binary image by To Binary.
4	Check your inbox for the Image preview email.	The email has the generated post text and an image attachment.
5	Open the approval email and choose Post it or Skip.	Post it continues; Skip ends the run without publishing.
6	If approved, check LinkedIn.	A post is published with image media.
7	Inspect the Posts sheet.	One row is appended with the topic, generated content, prompt, date, and available post ID.

Review the output before approval

- Check technical statements for accuracy. The workflow prompt attempts to restrict the draft to the supplied notes, but AI output still needs human review.
- Check whether the post reveals information you do not want public, including identifiers from labs, account details, or internal screenshots.
- Inspect the generated image for visual artifacts, accidental logos, or inappropriate technical depictions.
- Do not approve a post just because the workflow completed. Approval is a content-quality and safety checkpoint.

Approval timeout
The workflow configures a limited wait period in the Gmail Send and Wait node. If the timeout expires, treat the run as not approved and confirm the actual behavior in the execution log for your installed n8n version.

Chapter 9 | Troubleshooting

Symptom	Likely causes to check	Recommended action
Gemini node fails or returns quota error	API key, model availability, quota/rate limit, credential not assigned.	Test the credential; inspect the exact provider error; wait before retrying or select a model available under your account.
Image Prompt Agent expression error	Exported text and system message begin with ==.	Open the node fields and correct the expression marker to the syntax expected by the installed n8n version, then test the node alone.
Cloudflare returns 401/403	Wrong token, insufficient Workers AI permission, malformed Authorization header, or incorrect account ID.	Use a Workers AI token with required permission. Header value must use the form Bearer followed by one space and the token.
Cloudflare returns no usable image	Prompt/body issue, model response error, quota, or response shape differs from the Code node's expectations.	Inspect the HTTP Request output. Confirm the response contains an image value recognized by To Binary.
Image is missing in preview or LinkedIn	Binary data field mismatch or attachment configuration issue.	Check To Binary output for binary.data and the Email Preview attachment field. Confirm Restore Image1 returns the binary data.
Approval email not received	Wrong recipient, Gmail credential failure, spam filtering, send quota, or OAuth authorization issue.	Test the Gmail credential and preview node separately; confirm both nodes use the intended recipient.
LinkedIn authorization or publish error	Missing product/permission, callback mismatch, disconnected credential, token expiration, or unsupported account access.	Review the LinkedIn app configuration and the callback URL shown by n8n; reconnect and test with a simple post.
Google Sheets invalid_client	Wrong OAuth client ID/secret pairing, incorrect OAuth app type, redirect URI mismatch, or credential not configured.	Use matching client values; compare the redirect URI exactly; reconnect in n8n after correcting the OAuth client.
Google Sheets append fails or blank columns	Wrong spreadsheet ID, tab not named Posts, permission issue, or column-header mismatch.	Refresh the schema and ensure the headers match the node mapping exactly. Fix Auidence and the trailing space in Post.
The same topic can be submitted again	The current workflow has no actual duplicate lookup.	Add a Sheets read/search step before generation/publishing and branch on a documented duplicate comparison rule.

Debugging method

- Open the failed execution and find the first node that failed; later nodes may only be missing because that earlier node stopped the run.
- Run a single node using representative test data before rerunning the whole workflow. This reduces repeated API calls.
- Avoid repeated blind retries when using API plans with strict quotas. Read the exact provider error and retry only after addressing its cause.
- Keep a local, private copy of a known-good workflow export before changing expressions, credentials, or node mappings.

Chapter 10 | Security, Privacy, and Maintenance

Data handled by this workflow

Data	Where it is used or sent
Form topic and learning notes	Passed into the content-generation prompt sent to the configured Gemini model.
Profile, goals, and image style	Read by the agents from the Config node and sent as prompt context to Gemini when the relevant expressions are evaluated.
Generated image prompt	Sent to Cloudflare Workers AI in the image-generation request.
Post text and image	Emailed through Gmail for review; if approved, sent to LinkedIn to publish.
Run log values	Appended to the Posts worksheet in Google Sheets, including topic, audience, post, image prompt, date, and post ID where available.
OAuth tokens and API keys	Held in n8n credentials when properly configured; must not be copied into public workflow files or documentation.

Public repository hygiene

- Publish only a sanitized workflow JSON and dummy/sample data. Do not publish live credentials, token-bearing HTTP headers, private form URLs if you do not intend them to be public, or real private sheet identifiers and data.
- Review every screenshot for visible client secrets, personal email addresses, account IDs that should remain private, OAuth callback details, sheet IDs, execution data, or messages containing tokens.
- Keep production credentials only inside n8n credential storage. For additional security, restrict the n8n editor and protect backup/export files.
- If a key or secret has been exposed, revoke or rotate it, reconnect the relevant n8n credential, and review repository history so the old value is not left accessible in previous commits.

Privacy-policy consistency

The supplied index.html is a short privacy policy for the application. It describes use of LinkedIn account information, content creation/publishing, data sharing, retention, and contact. However, the actual workflow also sends submitted notes and configured profile context to Gemini, sends the generated image prompt to Cloudflare, routes preview/approval emails through Gmail, and logs content to Google Sheets. Before using the privacy page publicly for anyone other than yourself, revise it so these processing steps and the relevant third-party providers are described accurately. Do not make assurances about data retention or sharing unless you have verified the behavior and the providers' current terms.

Operational maintenance

- Periodically test every credential and reauthorize any expired or revoked OAuth connection.
- Review n8n and node version changes before upgrading a production workflow.
- Re-check provider quotas and current model availability when an API starts failing.
- Use versioned workflow backups before editing. Do not overwrite the only known-working export.
- Treat AI-generated content as a draft that needs factual and privacy review.

Credentials found in the supplied reference file
The Google Sheets setup reference PDF contains credential-like secret values. They are intentionally omitted from this manual. Rotate any exposed live client secret or API key before publicly releasing the repository or screenshots.

Chapter 11 | Known Limitations and Release Recommendations

Known limitations in the current exported workflow

Area	Current state	What to do before presenting it as production-ready
Duplicate prevention	Duplicate Check only validates form fields; it does not read Google Sheets.	Rename it to Form Validation or implement and test an actual sheet lookup and duplicate branch.
Sheet headers	The mapping uses Auidence and Post with a trailing space.	Normalize mapping/schema to Audience and Post, then run an append test.
Image Prompt Agent expression	Two prompt fields begin with == in the JSON.	Inspect and test the imported node; correct the leading marker if the expression fails.
Credential portability	Some credential references may not map to credentials in another n8n instance; several service credentials must be connected locally.	Document each credential type and require users to select their own credential after importing.
Provider quotas	Pauses exist between AI calls, but limits depend on provider account and model.	Add error handling/backoff where appropriate, avoid blind retry loops, and state that provider charges/limits may apply.
Human review	Image and post are emailed before publication, but review quality is still the user's responsibility.	Keep the approval step enabled and test both Post it and Skip branches.
Privacy statement	The HTML policy is broader and shorter than the full set of workflow data flows.	Update the policy to describe Gemini, Cloudflare, Gmail, LinkedIn, and Google Sheets processing.

Suggested release process

- Fix the sheet mapping mismatch and validate the Image Prompt Agent expressions in a fresh n8n instance.
- Re-import the cleaned JSON in a separate test workflow and connect all credentials from scratch.
- Test one low-risk topic from form submission through preview and approval. Verify the Skip branch separately.
- Verify that approval creates a LinkedIn post with the image, then check the Google Sheets row.
- Remove secrets and private identifiers from the PDF, README, HTML policy, workflow JSON, diagram, screenshots, and Git history.
- Update README claims to match actual behavior. Avoid describing duplicate prevention as active until it is implemented.
- Add a clear license and support/disclaimer section to the repository, and include the documentation PDF as the detailed setup guide.

Recommended project description

> *"LinkedIn Auto-Post Agent is a semi-automated n8n workflow that converts short learning notes into a draft LinkedIn post and AI-generated image, sends a preview for human approval, and publishes only after approval. It uses Google Gemini, Cloudflare Workers AI, Gmail, LinkedIn, and Google Sheets."*

Scope disclaimer

This is a practical project guide based on the supplied files. Provider dashboards, quotas, OAuth policies, and n8n interfaces may change. Confirm live provider requirements before use, and test on your own accounts before relying on the workflow.

Appendix A | Configuration Reference

Setting / credential	Where to configure it	Secret?	Notes
sheetId	Config node	No, but treat as private if the sheet is private.	Use the spreadsheet ID from the Google Sheets URL.
niche	Config node	No.	Current prompt expressions do not visibly use it directly.
myProfile	Config node	Potentially sensitive.	This content is sent to Gemini as prompt context.
myGoals	Config node	Potentially sensitive.	This content is sent to Gemini as prompt context.
imageStyle	Config node	Usually no.	Used to guide the image prompt.
cloudflareAccountId	Config node	Not a secret by itself.	Identifies the Cloudflare account in the API URL.
Gemini API credential	Gemini model nodes	Yes.	Keep key inside n8n credential storage.
Cloudflare Header Auth	Cloudflare Image1	Yes.	Authorization: Bearer <token>.
Gmail OAuth credential	Email Preview1 and Email Approval1	Yes.	Authorize the intended sending account.
LinkedIn credential	Post to LinkedIn	Yes.	Must be able to publish as the intended member.
Google Sheets OAuth credential	Save to Google Sheet	Yes.	Must have access to the target spreadsheet.

Suggested environment separation

Use separate test and production spreadsheets, credentials, and LinkedIn testing procedures wherever feasible. Avoid testing against an audience-visible account with a real publish unless you have reviewed the draft carefully. Keep the production workflow inactive until all nodes pass a controlled test.

Appendix B | Pre-Release Checklist

- ☐ The exported workflow JSON contains no real keys, client secrets, access tokens, or private credential data.
- ☐ Any credential-like secrets exposed in previous files or screenshots have been revoked or rotated.
- ☐ Both Gemini nodes are connected to a valid credential and a currently available model.
- ☐ Cloudflare account ID and Workers AI token are configured in the correct fields.
- ☐ Both Gmail nodes use the intended sending credential and approval recipient.
- ☐ The LinkedIn credential can publish as the intended user and uses the callback configured by n8n.
- ☐ The Google Sheet exists and the worksheet tab is named Posts.
- ☐ The Save to Google Sheet mapping and headers are standardized to Date, Topic, Audience, Post, ImagePrompt, LinkedInPostId.
- ☐ The Image Prompt Agent expression markers have been checked in the n8n editor.
- ☐ The actual duplicate-detection behavior is accurately described in README and documentation.
- ☐ One end-to-end test produces a preview email and an approval email.
- ☐ The Skip path is tested and does not publish.
- ☐ The Post it path publishes the approved image and text and adds one log row.
- ☐ The public privacy policy accurately describes data passed to Gemini, Cloudflare, Gmail, LinkedIn, and Google Sheets.
- ☐ Workflow diagram, screenshots, README, PDF, HTML, and repository history have been checked for private information.
- ☐ License and third-party usage terms are reviewed for the intended distribution and commercial use.

Ready to publish when

A fresh import can be configured from the written instructions, a test post succeeds only after approval, the sheet log is correct, and the repository contains no live credentials.

Appendix C | Terms and Reference Links

Term	Meaning in this project
n8n workflow	A set of connected nodes that execute a sequence of actions.
Credential	A saved authentication configuration used by an n8n node to access an external service.
OAuth 2.0	An authorization framework commonly used to let a user authorize an app without sharing their password with that app.
API key / token	A secret or bearer credential used to authenticate API calls; treat it like a password.
Binary data	File-like data handled separately from ordinary JSON in an n8n execution, such as the generated image.
Human-in-the-loop	A process where a person reviews the result and makes an explicit approval decision before an important action.
Rate limit / quota	A provider-enforced limit on requests or usage in a time window or billing period.

Reference resources

n8n documentation - https://docs.n8n.io/

Google AI Studio - https://aistudio.google.com/

Cloudflare dashboard - https://dash.cloudflare.com/

LinkedIn Developers - https://www.linkedin.com/developers/

Project files reviewed for this edition

Artifact	Use in this manual
README.md	Describes intended behavior, setup overview, tools, known limitations, and troubleshooting.
MyFinalAutomation_DUPLICATE(3).json	Source for the exported node list, node parameters, expressions, credentials references, and connections.
Duplicate n8n data - Google Sheets(1).pdf	Source for the parameter table and connection instructions. Credential-like values were deliberately not copied into this public-facing manual.
LinkedIn Auto-Post Workflow Diagram(1).png	Used as the architecture overview figure.
index.html	Reviewed for privacy-policy coverage and consistency with the actual workflow data flow.

End of manual.